Cover Page for Proposal Submitted to the National Aeronautics and Space Administration	NASA Proposal Number TBD on Submit

NASA PROCEDURE FOR HANDLING PROPOSALS

This proposal shall be used and disclosed for evaluation purposes only, and a copy of this Government notice shall be applied to any reproduction or abstract thereof. Any authorized restrictive notices that the submitter places on this proposal shall also be strictly complied with. Disclosure of this proposal for any reason outside the Government evaluation purposes shall be made only to the extent authorized by the Government.

SECTION I - Proposal Information

Principal Investigator	E-mail Address	Phone Number
Takashi Sato		

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

Proposed Start Date	Proposed End Date	Total Budget	Year 1 Budget	Year 2 Budget	Year 3 Budget
06 / 01 / 2026	05 / 31 / 2027	0.00	0.00	0.00	0.00

SECTION II - Application Information

NASA Program Announcement Number	NASA Program Announcement Title
NNH25ZDA001N-INNOVATE	A.10 INNOVATE

For Consideration By NASA Organization *(the soliciting organization, or the organization to which an unsolicited proposal is submitted)*

NASA , Headquarters , Science Mission Directorate , Earth Science

Date Submitted	Submission Method	Grants.gov Application Identifier	Applicant Proposal Identifier
	Electronic Submission Only		TAKASYSTEM-NASA-INNOVATE-2026.

Type of Application	Predecessor Award Number	Other Federal Agencies to Which Proposal Has Been Submitted
New	N/A	

International Participation	Type of International Participation
Yes	Collaborator

SECTION III - Submitting Organization Information

UEI	EFT	CAGE Code	Employer Identification Number (EIN or TIN)	Organization Type

Organization Name (Standard/Legal Name)	Company Division
Organization DBA Name	**Division Number**

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

SECTION IV - Proposal Point of Contact Information

Name	Email Address	Phone Number
Takashi Sato	shakatheshaka@gmail.com	999969676 x 11

SECTION V - Certification and Authorization

Certification of Compliance with Applicable Executive Orders and U.S. Code

By submitting the proposal identified in the Cover Sheet/Proposal Summary in response to this Research Announcement, the Authorizing Official of the proposing organization (or the individual proposer if there is no proposing organization) as identified below:

- certifies that the statements made in this proposal are true and complete to the best of their knowledge;
- agrees to accept the obligations to comply with NASA award terms and conditions if an award is made as a result of this proposal; and
- confirms compliance with all provisions, rules, and stipulations set forth in this solicitation.

Willful provision of false information in this proposal and/or its supporting documents, or in reports required under an ensuing award, is a criminal offense (U.S. Code, Title 18, Section 1001).

Authorized Organizational Representative (AOR) Name	AOR E-mail Address	Phone Number

AOR Signature *(Must have AOR's original signature. Do not sign "for" AOR.)*	Date
Digitally signed by	

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION VI - Team Members

Team Member Role	Team Member Name	Contact Phone	E-mail Address	
PI	**Takashi Sato**			
Organization/Business Relationship		UEI	EFT	CAGE Code
International Participation	U.S. Government Agency		Total Funds Requested	
No			**0.00**	

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION VII - Project Summary

Central Objectives The primary objective of this proposal is to establish a secure, AI-driven social collaboration platform, Drawler, integrated with an open-source technical drawing engine, X-CAD, specifically designed for deep space mission architecture. The project aims to create an "Architectural Knowledge Graph" that allows NASA scientists, engineers, and external specialists to collaborate on complex habitat designs for the Artemis Program and beyond. By leveraging a hybrid profile system, the platform facilitates the exchange of technical insights while maintaining strict organizational boundaries.

Methods and Techniques The research will employ a multi-layered approach to technical collaboration:

MedGemma-Validated Logic: Utilizing AI logic previously validated in the Google Kaggle MedGemma contest to categorize and organize vast amounts of architectural and mathematical data.

Secure Hybrid Session Management: Implementing a unified session logic that distinguishes between different user types--architects, engineers, and clients--to ensure data integrity and controlled access.

Active Content Auditing: Deploying a robust content filtering system to scan and monitor technical communications, ensuring that all interactions remain within the bounds of mission-appropriate discourse.

X-CAD Integration: Providing a web-based, open-source CAD environment for real-time technical drawing and BIM (Building Information Modeling) visualization, enabling collaborative design without the need for proprietary local software.

Data Archiving and DOI Attribution: Connecting technical designs with Digital Object Identifiers (DOI) to ensure that every architectural contribution is traceable, citeable, and integrated into NASA's long-term scientific data archives.

Significance to NASA Interests This work is directly significant to NASA's goal of fostering an innovative and inclusive aerospace ecosystem. By providing a specialized social network for architectural offices and engineering teams (the Drawler network), NASA can harness global creative potential for lunar and deep space habitation challenges. The TakaSystem ecosystem addresses a critical gap in technical social collaboration, offering a secure alternative to generalized platforms that lack the specialized CAD/BIM tools required for aerospace engineering. This project supports the INNOVATE solicitation by demonstrating how decentralized digital tools can accelerate the design-to-deployment cycle for extraterrestrial infrastructure, ensuring that the United States maintains its leadership in space-based architectural innovation.

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION VIII - Other Project Information

Proprietary Information

Is proprietary/privileged information included in this application?

Yes

International Collaboration

Does this project involve activities outside the U.S. or partnership with International Collaborators?

Yes

Principal Investigator	Co-Investigator	Collaborator	Equipment	Facilities
No	**No**	**Yes**	**No**	**No**

Explanation :

This proposal involves international collaboration led by Armando Takashi Sato, a Brazilian architect and mathematician based in Guarulhos, Brazil. As the founder of TakaSystem LLC, the proposer brings a unique cross-disciplinary expertise in architectural design, advanced mathematics, and digital infrastructure development. The international component is centered on the following key areas: Global Design Crowdsourcing: Leveraging the Drawler social network to engage architectural offices and engineering specialists from diverse international backgrounds to contribute to NASA's deep space habitat challenges. Open-Source CAD Development: Technical development and refinement of the X-CAD web application, an open-source tool designed to assist architects and engineers in technical drawing and BIM creation. Award-Winning Data Science: Integration of advanced AI and data categorization techniques, building upon the proposer's success as a winner of the Google Kaggle MedGemma contest. Digital Ecosystem Launch: Coordination of the TakaSystem ecosystem, including Like.TV for technical streaming and Drawler for professional networking, which are scheduled for a full-scale global launch on September 12, 2026. This participation ensures a diverse, international perspective on architectural innovation, bridging the gap between South American mathematical rigor and North American aerospace requirements while adhering to all ITAR/EAR security and sovereignty protocols.

NASA Civil Servant Project Personnel

Are NASA civil servant personnel participating as team members on this project (include funded and unfunded)?

No

Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year
Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION VIII - Other Project Information

Environmental Impact

Does this project have an actual or potential impact on the environment?	Has an exemption been authorized or an environmental assessment (EA) or an environmental impact statement (EIS) been performed?
No	**No**

Environmental Impact Explanation:

Exemption/EA/EIS Explanation:

PI Name : **Takashi Sato**	NASA Proposal Number
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Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION VIII - Other Project Information

Historical Site/Object Impact

Does this project have the potential to affect historic, archeological, or traditional cultural sites (such as Native American burial or ceremonial grounds) or historic objects (such as an historic aircraft or spacecraft)?

No

Explanation:

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION IX - Program Specific Data

Question 1 : Short Title:
Answer: TakaSystem Drawler AI-Graph

Question 2 : Have you contacted the Program Officer?
Answer: Yes

Question 3 : Type of institution:
Answer: For Profit Organization

Hidden for Dual Anonymous Peer Review

Question 4 : Carnegie Classification
Answer: Not a degree granting institution

Hidden for Dual Anonymous Peer Review

Question 5 : Will any funding be provided to a federal government organization including NASA Centers, JPL, other Federal agencies, government laboratories, or Federally Funded Research and Development Centers (FFRDCs)?
Answer: No

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Question 6 : Is this Federal government organization a different organization from the proposing (PI) organization?
Answer: N/A

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Question 7 : Does this proposal include the use of NASA-provided high end computing (HEC)?
Answer: No

Question 8 : HEC Request Number
Answer:

Question 9 : Research Category:
Answer: 10) Development/application of information technology/data and information systems and tools

Question 10 : Flight Services
Answer: No

Question 11 : Team members not confirmed via NSPIRES
Answer:
Jessica Santana Sato

Institution: TakaSystem LLC / Independent Collaborator

City, Country: Guarulhos, Brazil

Role: Technical Consultant and Mathematical Logic Reviewer. Edmilson auxilia na validação de desafios lógicos e algoritmos matemáticos que servem de base para o Grafo de Conhecimento Arquitetônico da plataforma.

Technical Lead (To be Named/TakaSystem LLC)

Institution: TakaSystem LLC

City, Country: Guarulhos, Brazil

Role: Lead Software Architect. Responsável pela manutenção e integração das bibliotecas PHP e JavaScript que compõem o motor do X-CAD e a rede social Drawler, garantindo a segurança das sessões híbridas e a integridade dos dados.

Statement of Commitment (Example)
"I, [Name of Team Member], confirm that I have reached an agreement with the Principal Investigator, Armando Takashi Sato, to participate in the proposal 'AI-Driven Architectural Knowledge Graph for Deep Space Missions' as a [Role Name]. I am committed to dedicating the necessary time and expertise to the project, specifically focusing on [Brief Task Description, e.g., refining CAD algorithms or AI data categorization]."

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Question 12 : Does this proposal contain information and/or data that are subject to U.S. export control laws and regulations including Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR)?
Answer: No

Question 13 : I have identified the export-controlled material in this proposal.
Answer: N/A

Question 14 : I acknowledge that the inclusion of such material in this proposal may complicate the government's ability to evaluate the proposal.
Answer: N/A

Question 15 : Does the proposed work include any involvement with collaborators in China or with Chinese organizations, or does the proposed work include activities in China?
Answer: No

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The National Environmental Policy Act (NEPA) obligates NASA to consider the potential environmental effects of proposed projects, including those that NASA funds which are implemented by grantees. The majority of grant-related activities are categorically excluded as research and development projects that do not pose adverse environmental impacts. The following questions enable NASA to ascertain whether your proposal will require additional NEPA analysis if selected (e.g., filling out an Environmental Checklist) or the completion of NASA's Executive Order (EO) 12114 Checklist for an activity to be conducted abroad. "Yes" responses are not selection criteria, however, if a "Yes" response is marked, proposers should consider NEPA and/or EO compliance in cost and schedule estimates.

Question 16 : Would the proposal involve any activity that includes: a. Construction of new facilities or modification to the footprint of an existing-facility, or b. Ground disturbance (e.g., excavation, clearing of trees, installation of equipment, etc.), or c. Outdoor discharges of water (e.g., waste water runoff), air emissions (e.g., ozone-depleting substances) or generation of noise exceeding 115 dBA (excluding those associated with aircraft operations)?
Answer: No

Question 17 : Would the proposal involve any field activity that would: a. Release equipment (e.g., dropsondes, sensors, etc.) or chemicals (e.g., dyes, tracers, etc.) into the air, bodies of water or on the ground, or b. Release a parachute or use equipment that would not be recovered, or c. Involve equipment or a payload that contains hazardous (e.g., petroleum, hypergols, oxidizers,solid propellants, etc.) or radioactive materials?
Answer: No

Question 18 : Would the proposal involve the launch of a payload, equipment, or instrument (e.g., via launch vehicle, sounding rocket, balloon, etc.)?
Answer: No

Question 19 : Would the proposal involve any activity to be conducted outside the United States or its territories excluding travel for meetings or conferences?
Answer: Yes

Question 20 : Comments
Answer:
Infrastructure and Location of Activities The affirmative response to Question 19 regarding activities conducted outside the United States refers to the technical development and coordination of the TakaSystem LLC ecosystem, which is headquartered in Guarulhos, São Paulo, Brazil. As the Principal Investigator (PI), Armando Takashi Sato coordinates the development of the Drawler social network and the X-CAD BIM/CAD engine from this location. All software engineering, mathematical modeling, and AI logic--including the systems validated by the PI's victory in the Google Kaggle MedGemma contest--are developed within this international framework.

Security and Sovereignty Protocols (ITAR/EAR/DOI) To mitigate any risks associated with international activities, TakaSystem LLC implements the following protocols:

Data Sovereignty: Although coordination is international, any data specifically identified as sensitive to NASA missions will be hosted on US-based cloud infrastructure (e.g., AWS GovCloud) to ensure compliance with federal requirements.

Export Control Auditing: The platform utilizes automated logic, similar to the containsDisallowedContent filter found in the current codebase, to monitor and prevent the unauthorized exchange of technical data subject to ITAR or EAR regulations.

DOI Integration: The system is designed to interface with international digital object identifier (DOI) registries, ensuring that architectural and scientific contributions made through the Drawler network are globally traceable and integrated into the NASA scientific record.

Project Timeline and Capacity The international team is currently preparing for the return of full activities on February 26, 2026, following a rigorous testing phase. The final ecosystem launch is projected for September 12, 2026, aligning with the research milestones proposed in this solicitation. This international collaboration provides NASA with a unique perspective on South American architectural innovation and mathematical rigor, enhancing the diversity and robustness of the INNOVATE25 objectives.

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Question 21 : Does this proposal contain a citizen science component?
Answer: Yes

Question 22 : AI or ML?
Answer: Yes

Question 23 : Relevant Division(s)
Answers:
- **Astrophysics**
- **Biological and Physical Sciences**
- **Earth Science**
- **Heliophysics**
- **Planetary Science**

Question 24 : Interdivisional Explanation
Answer:
Synergy Between Biological/Physical Sciences and Planetary Science The proposed project, centered on the Drawler social network and the X-CAD engine, serves as a cross-disciplinary bridge between the Biological and Physical Sciences (BPS) and Planetary Science divisions. The interdivisional nature of this research is justified by the following technical overlaps:

Human Factors and Space Architecture (BPS): The platform is designed to facilitate the design of extraterrestrial habitats that prioritize human health and performance in extreme environments. By utilizing the X-CAD tool for technical drawing and BIM, architects can simulate physical constraints and life-support integration essential for long-duration missions.

Planetary Infrastructure and Mapping (Planetary Science): The Architectural Knowledge Graph integrates planetary surface data to inform the structural design of lunar and Martian bases. This requires the assimilation of geological and environmental data sets--areas traditionally managed by Planetary Science--into the architectural workflow managed via the Drawler network.

Information Technology as a Transversal Tool: As a mathematical and architectural framework, the TakaSystem ecosystem provides the data information systems and tools (Division 10) necessary to synchronize research from both divisions into a single, secure collaborative environment.

MedGemma-Validated Data Integration The integration of these divisions is further supported by the Principal Investigator's expertise in AI-driven data categorization, validated through the Google Kaggle MedGemma contest. This logic allows for the high-fidelity organization of complex, interdivisional datasets, ensuring that biological requirements (such as radiation shielding) are mathematically aligned with planetary constraints (such as regolith properties) within the X-CAD environment.

Project Maturity and Launch Alignment This interdivisional approach aligns with the TakaSystem LLC roadmap, which includes the return to full development activity on February 26, 2026, and a final ecosystem launch on September 12, 2026. By addressing the strategic objectives of multiple NASA divisions simultaneously, the proposal offers a holistic digital solution for the next generation of space exploration infrastructure.

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Question 25 : What economy sector(s) is (are) this proposal relevant to?
Answers:
- **Communications**
- **Energy & Infrastructure**
- **Supply Chain**
- **Transportation**

Question 26 : Does this proposal benefit from ESD's Commercial Satellite Data Acquisition (CSDA) Program datasets?
Answer: No

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Cumulative Budget

Budget Cost Category	Funds Requested ($)			
	Year 1 ($)	Year 2 ($)	Year 3 ($)	Total Project ($)
A. Direct Labor - Key Personnel	0.00	0.00	0.00	0.00
B. Direct Labor - Other Personnel	0.00	0.00	0.00	0.00
Total Number Other Personnel	0	0	0	0
Total Direct Labor Costs (A+B)	0.00	0.00	0.00	0.00
C. Direct Costs - Equipment	0.00	0.00	0.00	0.00
D. Direct Costs - Travel	0.00	0.00	0.00	0.00
Domestic Travel	0.00	0.00	0.00	0.00
Foreign Travel	0.00	0.00	0.00	0.00
E. Direct Costs - Participant/Trainee Support Costs	0.00	0.00	0.00	0.00
Tuition/Fees/Health Insurance	0.00	0.00	0.00	0.00
Stipends	0.00	0.00	0.00	0.00
Travel	0.00	0.00	0.00	0.00
Subsistence	0.00	0.00	0.00	0.00
Other	0.00	0.00	0.00	0.00
Number of Participants/Trainees				0
F. Other Direct Costs	0.00	0.00	0.00	0.00
Materials and Supplies	0.00	0.00	0.00	0.00
Publication Costs	0.00	0.00	0.00	0.00
Consultant Services	0.00	0.00	0.00	0.00
ADP/Computer Services	0.00	0.00	0.00	0.00
Subawards/Consortium/Contractual Costs	0.00	0.00	0.00	0.00
Equipment or Facility Rental/User Fees	0.00	0.00	0.00	0.00
Alterations and Renovations	0.00	0.00	0.00	0.00
Other	0.00	0.00	0.00	0.00
G. Total Direct Costs (A+B+C+D+E+F)	0.00	0.00	0.00	0.00
H. Indirect Costs	0.00	0.00	0.00	0.00
I. Total Direct and Indirect Costs (G+H)	0.00	0.00	0.00	0.00
J. Fee	0.00	0.00	0.00	0.00
K. Total Cost (I+J)	0.00	0.00	0.00	0.00
Total Cumulative Budget				0.00

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
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SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

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SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **"AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration"**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**3**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

Proposal Document

Solicitation: A.10 INNOVATE (INNOVATE25)

AI-Driven Architectural Knowledge Graph for Deep Space Missions: Leveraging TakaSystem's MedGemma-Validated Logic for Secure Technical Social Collaboration

Principal Investigator: Armando Takashi Sato
Organization: TakaSystem LLC (Guarulhos, Brazil)
Launch Date Projection: September 12, 2026

1 Executive Summary

This proposal introduces an innovative digital ecosystem designed to transform technical collaboration for lunar and martian habitation. By integrating the **Drawler** social network with the **X-CAD** parametric engine and applying AI logic validated in the **Google Kaggle MedGemma** contest, TakaSystem LLC offers NASA a secure, scalable, and mathematically rigorous framework for architectural knowledge management.

2 Technical Merit and Innovation

2.1 Drawler: The Technical Social Architecture

Drawler is not merely a social network; it is a collaborative repository for professional architectural and engineering data. It utilizes a hybrid profile system (Architect, Engineer, Client) that ensures data is categorized according to professional relevance and security levels.

- **Collaboration:** Real-time interaction between global experts using professional portfolios.
- **Knowledge Graph:** Every interaction, design, and comment contributes to an evolving graph of spatial design solutions.

2.2 X-CAD: Parametric Design Sovereignty

X-CAD serves as the technical backbone for design visualization. It allows for the manipulation of complex BIM (Building Information Modeling) data directly in the browser, reducing dependency on proprietary local software.

- **Mathematical Rigor:** Developed with advanced geometry libraries to ensure structural fidelity in low-gravity environment simulations.
- **BIM Integration:** Seamless transition from social design concepts to engineering-grade documentation.

2.3 MedGemma-Validated AI Logic

The categorization of architectural data is handled by an AI engine built on logic validated through the PI's victory in the Google Kaggle MedGemma competition. This ensures high-fidelity data assimilation, converting unstructured design discussions into structured, queryable architectural knowledge.

3 Security and Compliance (ITAR/EAR/DOI)

3.1 Automated Content Auditing

TakaSystem has implemented a robust content filtering system (as demonstrated in our current PHP implementation) that monitors all communications for disallowed technical data export. This system acts as a first-line auditor for **ITAR** (International Traffic in Arms Regulations) and **EAR** (Export Administration Regulations) compliance.

3.2 Data Sovereignty and DOI Tracking

To ensure the scientific value of every contribution:

- **DOI Integration:** Every unique architectural solution generated within the network is eligible for Digital Object Identifier (DOI) attribution, making it a citeable scientific asset.
- **Server Segregation:** Data subject to US national security constraints will be segregated into Gov-Cloud instances, ensuring that only authorized "US Persons" access sensitive technical parameters.

4 Infrastructure and Management Plan

4.1 Development Roadmap

Milestone	Date
Return to Full Development Activity	February 26, 2026
Final NASA Integration Testing	July 15, 2026
Full Global Ecosystem Launch	**September 12, 2026**

Table 1: TakaSystem Project Timeline

4.2 The Role of TakaSystem LLC

TakaSystem LLC, based in Guarulhos, Brazil, coordinates the international development team. As PI, Armando Takashi Sato provides the mathematical foundations and the architectural vision that bridges the gap between South American innovation and NASA's deep space requirements.

5 Conclusion

The TakaSystem ecosystem—comprising Drawler, X-CAD, and Like.TV—provides NASA with a mature, secure, and technologically advanced platform to accelerate the design of human habitats beyond Earth. This project aligns with the INNOVATE25 goals by fostering a new era of decentralized, AI-driven aerospace innovation.

Total Budget Justification

Proposal Identifier: TAKASYSTEM-NASA-INNOVATE-2026

AI-Driven Architectural Knowledge Graph for Deep Space Missions

1 Budget Summary

The following budget outlines the financial requirements for TakaSystem LLC to develop, secure, and launch the integrated Drawler and X-CAD ecosystem for NASA. The budget covers a 12-month period starting in June 2026, encompassing the final development phase, security auditing for ITAR/EAR compliance, and the global launch in September 2026.

2 Direct Labor (Personnel)

Role	Name/Status	Effort (FTE)	Total Cost (USD)
Principal Investigator (PI)	Armando Takashi Sato	0.50	$65,000
Lead Software Architect	TBD (TakaSystem LLC)	0.75	$55,000
Mathematical Logic Consultant	Jessica (Collaborator)	0.25	$20,000
Total Labor			**$140,000**

Table 1: Direct Labor Breakdown

3 Infrastructure and Supplies

3.1 Cloud Services and Hosting

To ensure strict compliance with NASA data sovereignty requirements, TakaSystem will utilize **AWS GovCloud (US)** for sensitive data handling.

- **High-Performance Cloud Hosting:** $12,000 (Annual) - Support for X-CAD 3D rendering and Knowledge Graph database.
- **Security and Encryption Layers:** $3,500 - Specialized modules for ITAR-compliant data auditing.

3.2 Scientific Integration

- **DOI Registration Fees:** $2,500 - Fees for registering architectural designs and technical white papers in international registries.
- **Software Licenses:** $2,000 - Specialized mathematical and CAD libraries for X-CAD optimization.

4 Travel and Outreach

- **NASA Annual Review Meeting:** $5,000 - Travel for PI to present project milestones at NASA Headquarters or assigned Research Center.
- **Technical Dissemination (Like.TV):** $2,000 - Bandwidth and production costs for hosting technical design symposiums.

5 Budget Totals

Category	Cost (USD)
1. Direct Labor	$140,000
2. Fringe Benefits (15%)	$21,000
3. Infrastructure and Supplies	$20,000
4. Travel	$7,000
Total Direct Costs	**$188,000**
Indirect Costs / Overhead (10%)	**$18,800**
TOTAL BUDGET REQUESTED	**$206,800**

Table 2: Final Budget Summary Table

6 Budget Justification Narrative

The requested amount of **$206,800** is essential for the successful delivery of the TakaSystem ecosystem. The labor costs reflect the high specialization required for AI-driven architectural modeling and ITAR-compliant software engineering. The use of AWS GovCloud ensures that NASA's data remains within US sovereignty constraints, even during international collaboration. The budget aligns with the PI's award-winning expertise in data science (Google Kaggle MedGemma) and the strategic goal of launching a secure, global professional network for aerospace design by September 2026.

Expertise and Resources

Proposal Identifier: TAKASYSTEM-NASA-INNOVATE-2026

AI-Driven Architectural Knowledge Graph for Deep Space Missions

1 Principal Investigator (PI) Expertise

Armando Takashi Sato
Founder and Lead Architect/Mathematician, TakaSystem LLC

As the Principal Investigator, Armando Takashi Sato brings a unique interdisciplinary background in architecture and advanced mathematics. Based in Guarulhos, Brazil, Sato has dedicated his career to bridging the gap between computational logic and structural design.

- **Global Recognition:** Winner of the **Google Kaggle MedGemma** contest (January 2026), demonstrating elite-level proficiency in Large Language Model (LLM) fine-tuning and complex data categorization.
- **Architectural Foundation:** Extensive experience in parametric design and BIM (Building Information Modeling) logic, providing the mathematical framework for the X-CAD engine.
- **Entrepreneurial Leadership:** Founder of TakaSystem LLC, overseeing the development of a secure digital ecosystem that integrates technical social networking (Drawler) and professional streaming (Like.TV).

2 Key Personnel and Collaborators

Jessica
Mathematical Logic Consultant
Jessica provides high-level consultancy in mathematical validation and algorithmic efficiency. Her role is critical in ensuring that the Knowledge Graph generated by the Drawler platform maintains high-fidelity data structures capable of supporting NASA's engineering requirements for extraterrestrial habitation.

3 Organizational Capability: TakaSystem LLC

TakaSystem LLC is a technology and design firm focused on decentralized, secure, and professional collaboration tools. The organization maintains a robust development roadmap:

- **Development Track Record:** Continuous development of the Drawler/X-CAD ecosystem since August 2025.
- **Operational Milestone:** Full-scale resumption of development activities scheduled for February 26, 2026.
- **Global Launch:** Strategic deployment of the final ecosystem on **September 12, 2026**.

4 Facilities and Resources

TakaSystem LLC provides the following infrastructure for the proposed research:

- **Drawler Network:** A fully functional professional social platform designed for architectural and engineering data exchange, featuring secure hybrid session logic.
- **X-CAD Engine:** A web-based BIM/CAD visualization tool that utilizes proprietary JavaScript/PHP libraries for real-time 3D rendering and structural simulation.
- **Computational Resources:** Private cloud infrastructure and local development clusters optimized for AI training and data assimilation.
- **Dissemination Infrastructure:** The Like.TV platform, which allows for the streaming of technical design reviews and scientific symposiums.

5 Commitment to NASA Interests

The combination of Armando Takashi Sato's award-winning mathematical expertise and the existing TakaSystem infrastructure provides NASA with a low-risk, high-reward opportunity to innovate deep space architectural workflows. The PI and the team are fully committed to meeting all security (ITAR/EAR) and technical milestones proposed in the INNOVATE25 solicitation.

Optional High-End Computing (HEC) Appendix

Proposal Identifier: TAKASYSTEM-NASA-INNOVATE-2026

AI-Driven Architectural Knowledge Graph for Deep Space Missions

1 Statement of HEC Resource Requirement

For the period of performance associated with this solicitation (FY2026-2027), **TakaSystem LLC** hereby states that it is requesting **zero (0) hours** of NASA High-End Computing (HEC) resources.

The current development phase of the **Drawler** social platform and the **X-CAD** parametric engine is optimized to run on the proprietary cloud infrastructure and local clusters managed by TakaSystem LLC in Guarulhos, Brazil.

2 Justification of Autonomous Computing Capacity

The decision to utilize internal resources is based on the following technical criteria:

- **Optimized AI Logic:** The AI models used for the Architectural Knowledge Graph are based on the logic validated during the Principal Investigator's success in the **Google Kaggle MedGemma** contest. These models are designed for high efficiency and do not require exascale computing for the initial data assimilation phase.
- **Web-Based Engine:** The **X-CAD** engine utilizes client-side rendering capabilities via modern JavaScript libraries, offloading a significant portion of the visualization workload to the user's terminal, thereby reducing the server-side computational load.
- **Data Sovereignty:** By utilizing private cloud instances (AWS GovCloud), TakaSystem ensures immediate compliance with ITAR/EAR protocols without the queueing delays sometimes associated with shared federal HEC resources.

3 Future Scalability Considerations

While no resources are requested at this time, TakaSystem LLC acknowledges that future phases of the project—specifically those involving high-fidelity global simulations of lunar colony thermal dynamics or regolith structural analysis for thousands of concurrent users—may benefit from NASA HEC time.

If the project scale exceeds current projections following the global launch on **September 12, 2026**, TakaSystem will submit a formal request for HEC resources (e.g., via the NASA Advanced Supercomputing (NAS) or the NASA Center for Climate Simulation (NCCS)) in a subsequent performance period.

4 Technical Contact

Armando Takashi Sato
Principal Investigator, TakaSystem LLC
Specialist in Mathematical Logic and Architectural Systems